

CCSU BCA Semester 2 – Object Oriented Programming

2026 Exam-Oriented Important Topics + Short Notes

1. OOP Basics

Short Note: Object-oriented programming organizes software around objects containing data and behavior. Main concepts are class, object, encapsulation, abstraction, inheritance and polymorphism.

Exam focus: Definition + key points + example/diagram/program where applicable.

2. Class & Object

Short Note: A class is a blueprint defining data members and methods. An object is an instance of a class. Learn declaration, creation and member access.

Exam focus: Definition + key points + example/diagram/program where applicable.

3. Encapsulation

Short Note: Encapsulation bundles data and methods and controls access to internal state. Access modifiers are commonly used to implement it.

Exam focus: Definition + key points + example/diagram/program where applicable.

4. Inheritance

Short Note: Inheritance allows a new class to reuse/extend an existing class. Learn base/derived class, common inheritance forms and advantages.

Exam focus: Definition + key points + example/diagram/program where applicable.

5. Polymorphism

Short Note: Polymorphism means one interface/name can represent different behavior. Learn compile-time overloading and run-time overriding concepts.

Exam focus: Definition + key points + example/diagram/program where applicable.

6. Abstraction

Short Note: Abstraction hides unnecessary implementation details and exposes essential behavior. Learn abstract classes/interfaces according to the language used in your course.

Exam focus: Definition + key points + example/diagram/program where applicable.

7. Constructors & Destructors

Short Note: A constructor initializes an object. A destructor/finalizer concept is used for cleanup depending on the language. Learn syntax and purpose for your prescribed language.

Exam focus: Definition + key points + example/diagram/program where applicable.

8. Exception Handling

Short Note: Exception handling manages abnormal situations without abruptly terminating normal program logic. Learn try/catch/throw or the equivalent constructs in the prescribed language.

Exam focus: Definition + key points + example/diagram/program where applicable.

■ Important Questions

- Explain **OOP Basics** with suitable example/diagram/formula/program.
- Explain **Class & Object** with suitable example/diagram/formula/program.
- Explain **Encapsulation** with suitable example/diagram/formula/program.
- Explain **Inheritance** with suitable example/diagram/formula/program.
- Explain **Polymorphism** with suitable example/diagram/formula/program.
- Explain **Abstraction** with suitable example/diagram/formula/program.
- Explain **Constructors & Destructors** with suitable example/diagram/formula/program.
- Explain **Exception Handling** with suitable example/diagram/formula/program.

Note: These are revision notes aligned to the CCSU BCA curriculum structure available for 2025–26. Check your college's paper combination/electives before the exam.